



Introducing the capstone project

Hi there. I'm so glad you're joining me for this last part of the program. This is an end-of-certificate project that Coursera and other learning platforms usually call a capstone.

The capstone brings everything you've learned together. You'll have the opportunity to take all of your new knowledge and put it into practice with a data analytics case study.

In this video, we'll talk more about what that entails and how it can help you stand out during a job search. Case studies are practice data analytics projects.

When you're job hunting, you might be asked to do a case study after the pre-screen call or the first interview. The case study is a common way for employers to assess job skills and gain insight into how you approach common data-related challenges. Different employers might send you different kinds of case studies. For example, you might be asked to clean and analyze a data set, offer a proposal around how to measure the success of a project, or figure out and define metrics of success for a specific product.

Usually, there's a time limit for the case study you've been asked to do. For example, a potential employer might give you some sample data and project questions and ask you to create a presentation or memo with your recommendations in 24 to 48 hours. That time limit can be a little challenging.

But the good news is, your answer to the case study doesn't have to be perfect. What's important is that you show off your thought process so that the interviewers can understand how you approach the problem. You can use the data analysis process we've learned throughout this program to guide you. Let's check out an example and break down all of the parts. This case study has all of the information we'd need to perform this task. It starts here, with the title and the industry focus: predicting employee attrition rate for human resources. It also includes a problem statement outlining what the overall goal is.

In this case, they're asking for a deep dive into key data analytics concepts to predict the employee attrition rate in the organization, and which factors influence an employee to leave the organization. Basically, this case study is interested in predicting the rate at which employees might leave the organization and why. There's some more specific goals in the next section. It's asking us to find the probability of an employee leaving the company over the next five years. That's pretty straightforward, but they're also interested in ways to improve employee retention. This next section is really key. The deliverables are what we'll actually give them once we've completed the case study.

In this example, they're asking for a presentation outlining our findings and recommendations. Finally, they've included some sections about the data we'll use for this task. Here, it's a dataset that we can download. Now we know more about case studies and how they might be presented to us in a job application process. But people who are passionate about data analytics will sometimes do case studies on their own time and add them to a personal portfolio. A portfolio is a collection of case studies that

can be shared with potential employers. Portfolios can be stored on public websites like GitHub, Kaggle, or Tableau, or on your blog.

Your portfolio can also be linked in your resume. This will give you examples of how you approached data tasks in the past that you can talk about in your interview. These portfolios showcase your skills and help you stand out in job applications. On top of the case study, we'll talk about building your portfolio and how to share it. This will be a great building block that you can use to build up your resume. Coming up, we'll check out some great examples of case studies and portfolios that will hopefully inspire you as you start your own.

Case study

A common way for employers to assess job skills and gain insight into how you approach common data related challenges

Portfolio

Collection of case studies that can be shared with potential employers

Example Data Analytics Case Study

Title *	Predicting Employee Attrition Rate
Industry focus *	HR
Problem statement *	Deep dive into key data analytics concepts to predict the employee attrition rate in the organization, as well as which factors influence an employee to leave the organization.
Business use case (what are you solving for?)	1. Predicting individual employee attrition 2. Depicting important factors for an employee to leave the organization 3. Improving employee retention
Goals / metrics	Find the probability of an employee leaving the firm for the next 5 years and identify how to improve employee retention
Deliverables	A presentation outlining your finding and recommendations
Are datasets available?	Yes
Dataset list	The data set can be downloaded from the link given in the below section
Websites to scrape the data needed *	https://www.kaggle.com/jacksonchou/hr-data-for-analytics